



iDPP@CLEF 2023: The Intelligent Disease Progression Prediction Challenge

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Abstract. *Amyotrophic Lateral Sclerosis (ALS)* and *Multiple Sclerosis (MS)* are chronic diseases characterized by progressive or alternate impairment of neurological functions (motor, sensory, visual, cognitive). Patients have to manage alternated periods in hospital with care at home, experiencing a constant uncertainty regarding the timing of the disease acute phases and facing a considerable psychological and economic burden that also involves their caregivers. Clinicians, on the other hand, need tools able to support them in all the phases of the patient treatment, suggest personalized therapeutic decisions, indicate urgently needed interventions.

The goal of iDPP@CLEF is to design and develop an evaluation infrastructure for AI algorithms able to:

1. better describe disease mechanisms;
2. stratify patients according to their phenotype assessed all over the disease evolution;
3. predict disease progression in a probabilistic, time dependent fashion.

iDPP@CLEF run as a pilot lab in CLEF 2022, offering tasks on the prediction of ALS progression and a position paper task on explainability of AI algorithms for prediction; 5 groups submitted a total of 120 runs and 2 groups submitted position papers.

iDPP@CLEF will continue in CLEF 2023, focusing on the prediction of MS progression and exploring whether pollution and environmental data can improve the prediction of ALS progression.

1 Introduction

Amyotrophic Lateral Sclerosis (ALS) and *Multiple Sclerosis (MS)* are severe chronic diseases characterized by a progressive but variable impairment of neurological functions, characterized by high heterogeneity both in presentation features and rate of disease progression. As a consequence patients' needs are different, challenging both caregivers and clinicians. Indeed, the time of relevant events is variable, which is associated with uncertainty regarding the opportunity of critical interventions, like non-invasive ventilation and gastrostomy in the case of ALS, with implications on the quality of life of patients and their caregivers. For this reason, clinicians need tools able to support their decision in all phases of disease progression and underscore personalized therapeutic decisions. Indeed, this heterogeneity is partly responsible for the lack of effective prognostic tools in medical practice, as well as for the current absence of a therapy able to effectively slow down or reverse the disease course. On the one hand, patients need support for facing the psychological and economic burdens deriving from the uncertainty of how the disease will progress; on the other, clinicians require tools that may assist them throughout the patient's care, recommending tailored therapeutic decisions and providing alerts for urgently needed actions.

We need to design and develop *Artificial Intelligence (AI)* algorithms to:

- stratify patients according to their phenotype all over the disease evolution;
- predict the progression of the disease in a probabilistic, time dependent way;
- better describe disease mechanisms.

The *Intelligent Disease Progression Prediction at CLEF (iDPP@CLEF)* lab¹ aims to deliver an evaluation infrastructure for driving the development of such AI algorithms. Indeed, in this context, it is fundamental, even if not so common yet, to develop shared approaches, promote the use of common benchmarks, foster the comparability and replicability of the experiments. Differently from previous challenges in the field, iDPP@CLEF addresses in a systematic way some issues related to the application of AI in clinical practice in ALS and MS. In addition to defining risk scores based on the probability of occurrence of an event in the short or long term period, iDPP@CLEF also addresses the issue of providing information in a more structured and understandable way to clinicians.

The paper is organized as follows: Section 2 discusses related works; Sect. 3 presents what has been done in iDPP@CLEF 2022 while Sect. 4 introduces the plans for iDPP@CLEF 2023; finally, Sect. 5 draws some conclusions.

2 Related Works

Within CLEF, there have been no other labs on this or similar topics before.

Outside CLEF, there have been a recent challenge on Kaggle² in 2021 and some older ones, the DREAM 7 ALS Prediction challenge³ in 2012 and the

¹ <https://brainteaser.health/open-evaluation-challenges/>.

² <https://www.kaggle.com/alsgroup/end-als>.

³ <https://dreamchallenges.org/dream-7-phil-bowen-als-prediction-prize4life/>.

DREAM ALS Stratification challenge⁴ in 2015. The Kaggle challenge used a mix of clinical and genomic data to seek for insights about the mechanisms of ALS and difference between people with ALS who progress faster versus those who develop it more slowly. The DREAM 7 ALS Prediction challenge [8] asked to use 3 months of ALS clinical trial information (months 0–3) to predict the future progression of the disease (months 3–12), expressed as the slope of change in *ALS Functional Rating Scale Revisited (ALSFRS-R)* [4], a functional scale that ranges between 0 and 40. The DREAM ALS Stratification challenge asked participants to stratify ALS patients into meaningful subgroups, to enable better understanding of patient profiles and application of personalized ALS treatments. Differently from these previous challenges, iDPP@CLEF focuses on explainable AI and on temporal progression of the disease.

Finally, when it comes to MS, studies are mostly conducted on closed and proprietary datasets and iDPP@CLEF represents one of the first attempts to create a public and shared dataset.

3 iDPP@CLEF 2022

iDPP@CLEF run as a pilot lab for the first time in CLEF 2022⁵ [6, 7] and focused on pilot activities aimed both at an initial exploration of ALS progression prediction and at understanding of the challenges and limitations to refine and tune the labs itself for future iterations.

Tasks. iDPP@CLEF 2022 consisted of the following tasks:

- **Pilot Task 1 - Ranking Risk of Impairment:** it focused on ranking of patients based on the risk of impairment in specific domains. More in detail, we used the ALSFRS-R scale [4] to monitor speech, swallowing, handwriting, dressing/hygiene, walking and respiratory ability in time and asked participants to rank patients based on time to event risk of experiencing impairment in each specific domain.
- **Pilot Task 2 - Predicting Time of Impairment:** it refined Task 1 by asking participants to predict when specific impairments will occur (i.e. in the correct time-window). In this regard, we assessed model calibration in terms of the ability of the proposed algorithms to estimate a probability of an event close to the true probability within a specified time-window.
- **Position Paper Task 3 - Explainability of AI algorithms:** we evaluated proposals of different frameworks able to explain the multivariate nature of the data and the model predictions.

Participation. 43 participants registered for iDPP@CLEF 2022 and 5 participants successfully submitted a total of 120 runs for Task 1 and Task 2; moreover, 2 position papers were submitted for the explainability task, as detailed in Table 1. Submission of participants are openly available in git repositories⁶.

⁴ <https://dx.doi.org/10.7303/syn2873386>.

⁵ <https://brainteaser.health/open-evaluation-challenges/idpp-2022/>.

⁶ <https://bitbucket.org/brainteaser-health/>.

Table 1. Break-down of the runs submitted by participants for each task and sub-task. Participation in Task 3 does not involve submission of runs and it is marked just with a tick.

Team Name	Total	Task 1	Task 2	Task 3	Paper
BioHIT	18	9	9	–	–
CompBioMed	40	22	18	–	Pancotti et al. [12]
FCOOL	15	–	15	✓	Branco et al. [2] and Nunes et al. [11]
LIG GETALP	23	12	11	–	Mannion et al. [10]
SBB	24	12	12	–	Trescato et al. [13]
UNIPV	–	–	–	✓	Buonocore et al. [3]
Total	120	55	65		

Datasets. iDPP@CLEF 2022 created 3 datasets, for the prediction of specific events related to ALS, consisting of fully anonymized data from 2,250 real patients from medical institutions in Turin, Italy, and Lisbon, Portugal. The datasets contain both static data about patients, e.g. age, onset date, gender, ... and event data, i.e. 18,512 ALSFRS-R questionnaires and 4,015 spirometries.

The following data are available for both the training and the test sets:

- the first available ALSFRS-R questionnaire at **Time 0** (both single question scores and total score). , for example, time-of-onset and time-of-diagnosis are expressed as relative delta with respect to **Time 0** in months (also fractions);
- the slope of the ALSFRS-R score between time-of-onset and **Time 0** as:

$$slope = \frac{48 - \text{ALSFRS-R-score}(\text{Time0})}{\text{Time0} - \text{TimeOnset}}$$

- all the other static data, with a complete list available at <http://brainteaser.dei.unipd.it/challenges/idpp2022/assets/other/static-vars.txt>
- visits, containing either other ALSFRS-R questionnaires or Spirometry, i.e. *Forced Vital Capacity (FVC)*. The complete list of variables for each visit is available at <http://brainteaser.dei.unipd.it/challenges/idpp2022/assets/other/visits.txt>.

Measures. iDPP@CLEF adopted several state-of-the-art evaluation measures to assess the performance of the prediction algorithms, among which:

- *ROC curve and/or the precision-recall curve (and area under the curve)* to show the trade-off between clinical sensitivity and specificity for every possible cut-off of the risk scores;
- *Concordance Index (C-index)* to summarize how well a predicted risk score describes an observed sequence of events.
- *E/O ratio and Brier Score* to assess whether or not the observed event rates match expected event rates in subgroups of the model population.

- *Specificity and recall* to assess, for each interval, the ability of the models of correctly identify true positives and true negatives.
- *Distance* to assess how far the predicted time interval was from the true time interval.

Approaches. BoiHIT explored the use of logistic regression, random forest classifiers, XGBoost, and LightGBM. Decision trees and boosting approaches were preferred due to their ability to deal with both categorical and numerical/continuous features and the interpretability they offer. Even if LightGBM was the model with the best performance, BoiHIT found out that this kind of approaches might not be appropriate for time dependent problems and that time to event analysis methods, such as survival analysis, might yield better results.

CompBioMed [12] considered three main approaches. The simplest one consisted on fitting a standard survival predictor separately for each event as outlined above for independent events, called Naive Multiple Event Survival (NMES). Another was the recently developed Deep Survival Machine (DSM), based on deep learning and capable of handling competing risks. Finally, they also proposed a time-aware classifier ensemble method, that also handles competing risks, called Time-Aware Classifier Ensemble (TACE). All the above approaches achieved comparable performance among them.

FCOOL [2] proposes a hierarchical approach, with a first-stage event prediction, followed by specialized models predicting the time window to a particular event. The procedure is three-fold: first, it creates patient snapshots based on clustering with constraints, thus organizing patient records in an efficient manner. Second, it uses a pattern-based approach that incorporates recent advances on temporal pattern mining to the context of classification. This approach performs end-stage event prediction while allowing the entire patient’s medical history to be considered. Finally, exploiting the predictions from the previous step, specialized models are learned using the original features to predict the time window to an event. This two-stage prediction approach aimed to promote homogeneity and lessen the impact of class imbalance, in comparison to performing one single multilabel task.

LIG GETALP [10] employed Cox’s proportional hazards model to the task of ranking the risk of impairment, using the gradient boosting learning strategy. The output of the time-independent part of the survival function calculated by the gradient boosting survival analysis method is then mapped to the interval $(0, 1)$, via a sigmoid function. To estimate the time-to-event, LIG GETALP used a regression model based on Accelerated Gradient Boosting (AGB). This being a standard regression model, it does not take censoring into account and [10] uses class predictions based on the Task 1 survival model to “censor” the time-to-event predictions.

SBB [13] considered three survival analysis methods, namely: Cox, SSVM, and RSF. They were chosen to represent a broad spectrum of baseline models including parametric (SSVM), semiparametric (Cox), linear (Cox, SSVM), and nonlinear (RSF) models.

For task 3, Nunes et al. [11] proposed a novel approach that generates semantic similarity-based explanations for patient-level predictions. The underlying idea is to explain the prediction for one patient by considering aspect-oriented semantic similarity with other relevant patients based on the most important features used by ML approaches or selected by users. To build rich and easy to understand semantic-similarity based explanations, [11] developed five steps: (1) the enrichment of the Brainteaser Ontology [1] through integration of other biomedical ontologies; (2) the semantic annotation of patients (if not already available); (3) the similarity calculation between patients; (4) selection of the set of patients to explain a specific prediction; and (5) the visualization of the generated similarity-based explanations.

For task 3, Buonocore et al. [3] trained a set of 4 well-known classifiers to predict death occurrence: Gradient Boosting (using XGB implementation), Random Forest, Logistic Regression and Multilayer perceptron. For the *eXplainable AI(XAI)* methods [3] focused our attention on three different methods for post-hoc, model-agnostic, local explainability, selecting SHAP, LIME and AraucanaXAI. Then, [3] evaluated and compared XAI approaches in terms of a set of metrics defined in previous research on XAI in healthcare: *identity*: if there are two identical instances, they must have the same explanations; *fidelity*: concordance of the predictions between the XAI surrogate model and the original ML model; *separability*: if there are 2 dissimilar instances, they must have dissimilar explanations; *time*: average time required by the XAI method to output an explanation across the entire test set.

4 iDPP@CLEF 2023

iDPP@CLEF 2023⁷ will organize the following activities:

- **Task 1 – Predicting Risk of Disease Worsening (MS)**: It focuses on ranking subjects based on the risk of worsening, setting the problem as a survival analysis task. More specifically the risk of worsening predicted by the algorithm should reflect how early a patient experiences the event “worsening” and should range between 0 and 1. Worsening is defined on the basis of the *Expanded Disability Status Scale (EDSS)* [9], accordingly to clinical standards. In particular, we consider two different definitions of worsening corresponding to two different sub-tasks:
 - *Subtask 1a*: the patient crosses the threshold $EDSS \geq 3$ at least twice within one year interval;
 - *Subtask 1b*: the second definition of worsening depends on the first recorded value accordingly to current clinical protocols. If Baseline $EDSS < 1$, worsening event occurs when and increase of $EDSS$ by 1.5 points is first observed; if $1 \leq \text{Baseline } EDSS < 5.5$, worsening event occurs when and increase of $EDSS$ by 1 point is first observed; if baseline $EDSS \geq 5.5$, worsening event occurs when and increase of $EDSS$ by 0.5 points is first observed.

⁷ <https://brainteaser.health/open-evaluation-challenges/idpp-2023/>.

In both cases the occurrence of the worsening event and the time of occurrence will be pre-computed by the challenge organizers.

- **Task 2 – Predicting Cumulative Probability of Worsening (MS):** it refines Task 1 asking participants to explicitly assign the cumulative probability of worsening at different time windows, i.e. between years 2 and 4, 2 and 6, 2 and 8, 2 and 10. Worsening will be defined in two different ways in subtasks 2a and 2b similarly to Task 1.
- **Position Paper Task 3 – Impact of Exposition to Pollutants (ALS):** we will evaluate proposals of different approaches to assess if exposure to different pollutants is a useful variable to predict time to Percutaneous Endoscopic Gastrostomy (PEG), Non-Invasive Ventilation (NIV) and death in ALS patients.

We will provide retrospective, fully anonymized MS and ALS clinical data including demographic and clinical characteristics, coming from clinical institutions in Italy, Portugal, and Spain.

For Task 1 and Task 2 we will release a brand new dataset with MS data consisting of about 1,800 patients. Accordingly to the survival analysis settings, for each subject in the training we will provide a label 0 or 1 indicating if the subject experienced the event “worsening” (label 1) or not (label 0) and the time-of-event, which indicates the time of the event for subjects experiencing it, or the time of censoring if the patient has not experience the event yet at the time of data dump.

For Position Paper Task 3 we will re-use the ALS dataset developed in iDPP@CLEF 2022, consisting of about 2,250 patients, and will extend it with environmental and pollution data.

5 Conclusions

iDPP@CLEF is a new shared tasks focusing on predicting the temporal progression of ALS and MS and on the explainability of the AI algorithms for such prediction. The first edition focused on ALS progression prediction and participation was satisfactory, hinting at the interest of the community concerning the task. More so, the solutions identified by participants range over several different techniques and provided valid input to such a highly relevant domain as the prediction of the ALS progression.

For the second iteration, iDPP@CLEF 2023 we plan to investigate MS progression prediction and how to exploit pollution and environmental data to improve progression prediction of ALS.

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